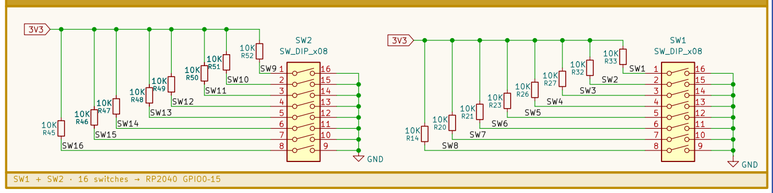
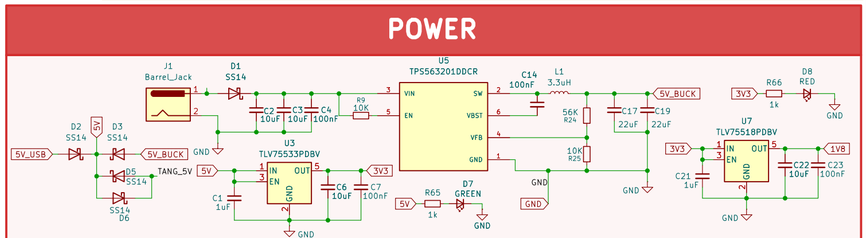
## DIP SWITCHES



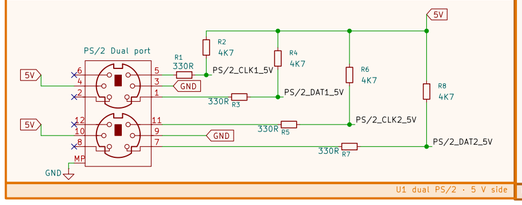

SW1 + SW2 · 16 switches → RP2040 GPIO0-15

## POWER



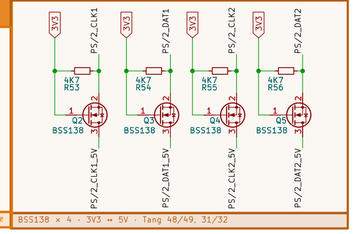

J1 6-17 V → buck 5.07 V → 5V → LDO 3V3 / 1V8

## PS/2



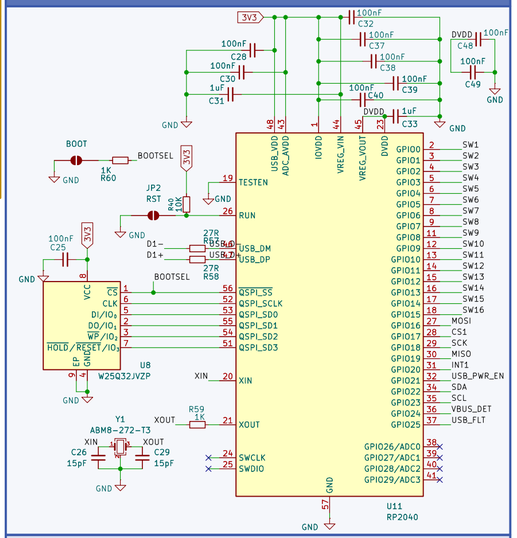

U1 dual PS/2 · 5 V side

## PS/2 LEVEL SHIFTER



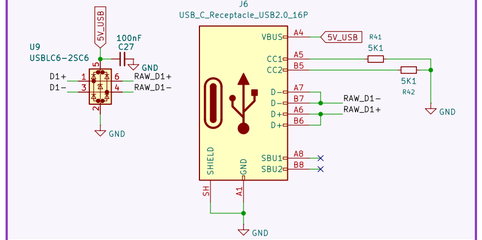

BSS138 × 4 · 3V3 ↔ 5V · Tang 48/49, 31/32

## RP2040



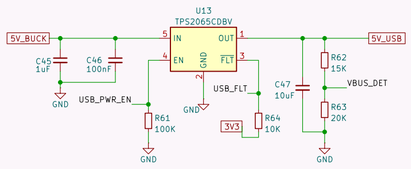

W25Q32 flash · 12 MHz · SPI0 slave · I2C1 → J3

## USB-C



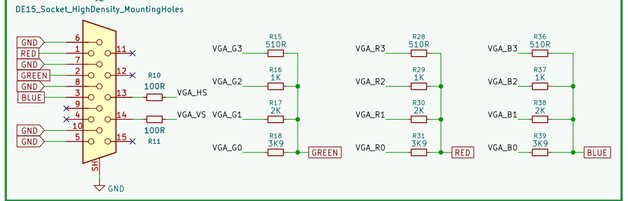

J6 · CC 5k1 · USBLC6 ESD · 27R → RP2040

## USB HOST POWER



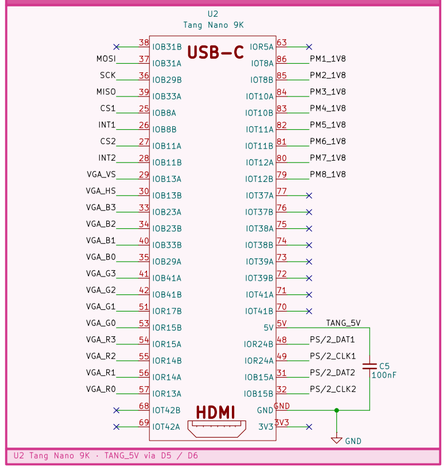

TPS2065C 5V_BUCK → 5V_USB · EN GPIO21 · FLT GPIO25 · DET GPIO24

## VGA



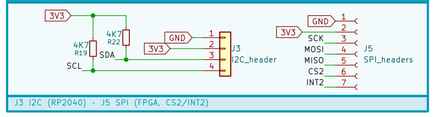

R-DAC 4 bit × RGB · HS 30 · VS 29 · J2 DE15

## TANG NANO



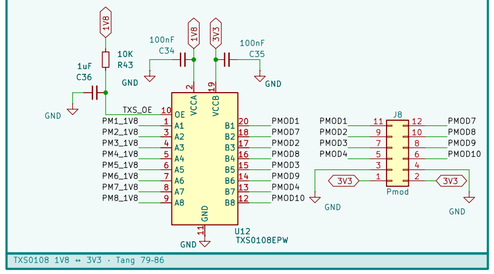

U2 Tang Nano 9K · TANG_5V via D5 / D6

## EXTENSION HEADERS




J3 I2C (RP2040) · J5 SPI (FPGA, CS2/INT2)

## PMOD

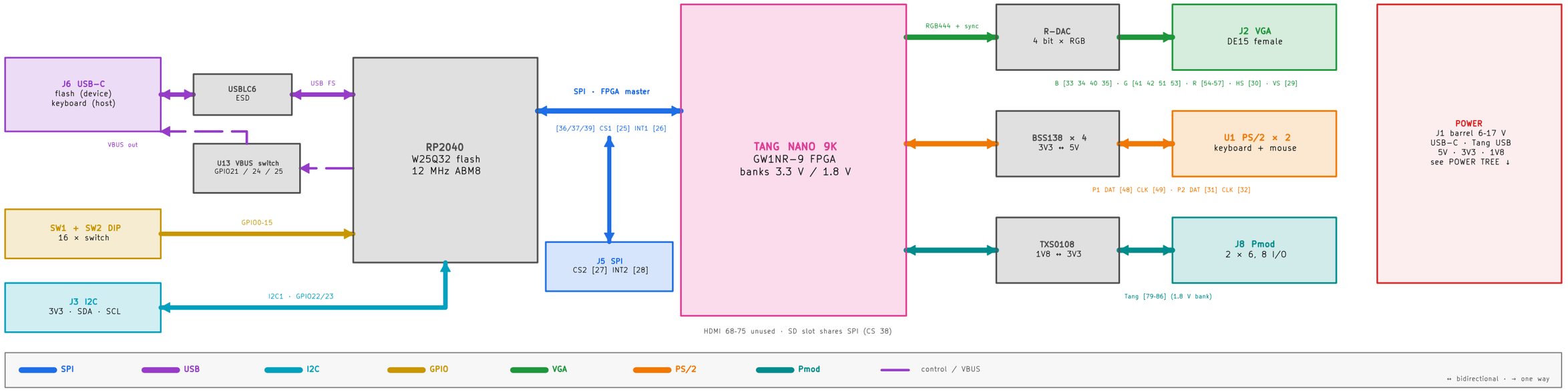



TXS0108 1V8 ↔ 3V3 · Tang 79-86

| | | |
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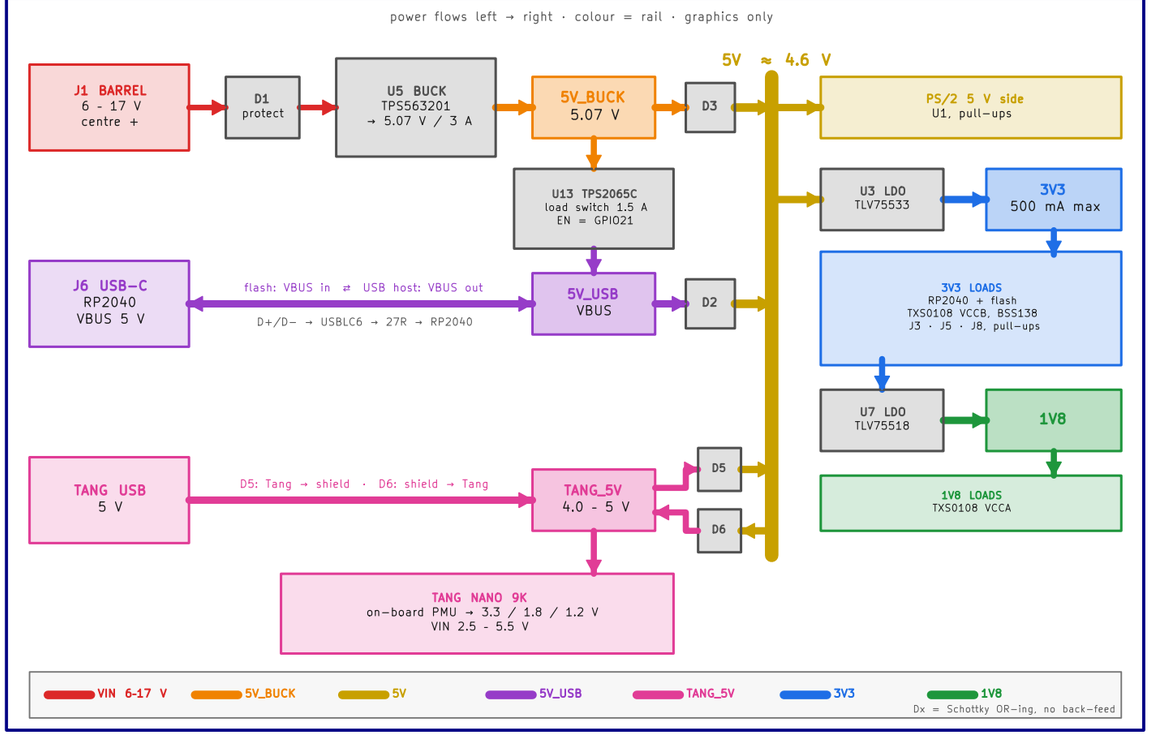
# BLOCK DIAGRAM

signal flow · colour = interface · Tang pin numbers in brackets




# POWER TREE

power flows left → right · colour = rail · graphics only




Sheet: /block-diagram/
File: block_diagram.kicad_sch

**Title:**

| Size: A3 | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 1/2 |